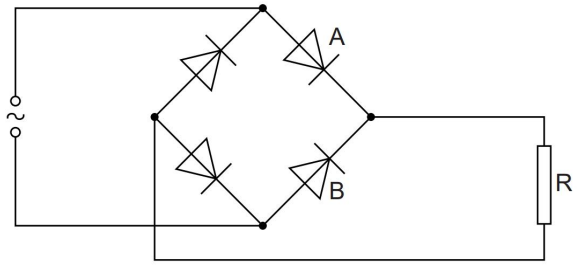
21.2 Exercises - Rectification and smoothing

1. A sinusoidal alternating voltage is to be rectified.

a. Suggest one advantage of full-wave rectification as compared with half-wave rectification.

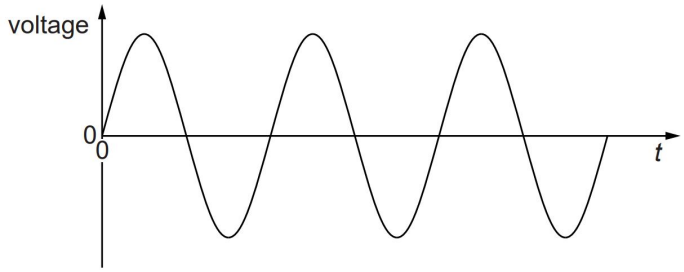
.....[1]

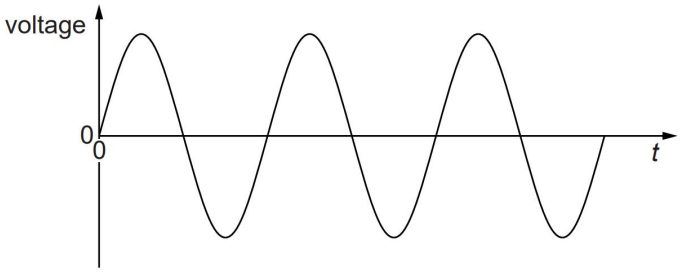
b. The rectification is produced using the circuit of Fig.



All the diodes may be considered to be ideal.

The variation with time t of the alternating voltage applied to the circuit is shown in Fig. below.





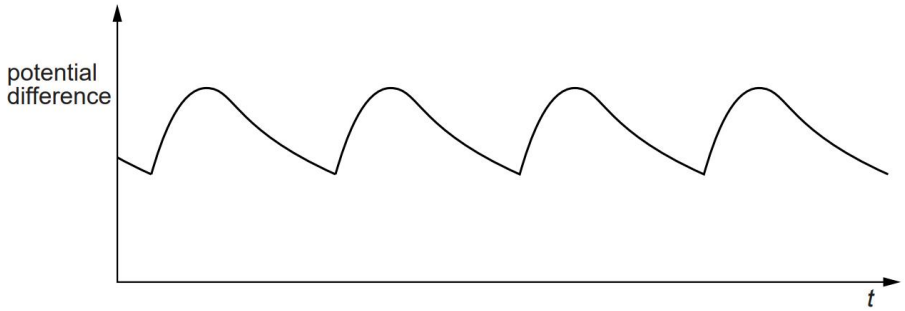
i. On the axes of Fig. left, draw a graph to show the variation with time t of the potential difference across diode A. [1]

ii. On the axes of Fig. right, draw a graph to show the variation with time t of the potential difference across diode B. [1]

c.

i. On Fig.1, draw the symbol for a capacitor, connected into the circuit so as to provide smoothing. [1]

ii. Fig. below shows the variation with time t of the smoothed potential difference across the resistor R in Fig.1.



a) State how the amount of smoothing may be increased.

.....[1]

b) On Fig., draw the variation with time t of the potential difference across resistor R for increased smoothing. [2]

2. The output potential difference (p.d.) of an alternating power supply is represented by

$$V = 320sin(100\pi t)$$

where V is the p.d. in volts and t is the time in seconds.

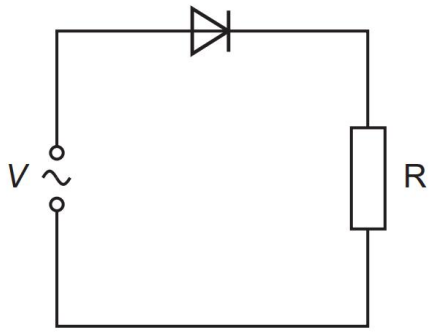
a. Determine the root-mean-square (r.m.s.) p.d. of the power supply.

$V_{r.m.s.}$ = ... V [1]

b. Determine the period T of the output.

T = ... s [2]

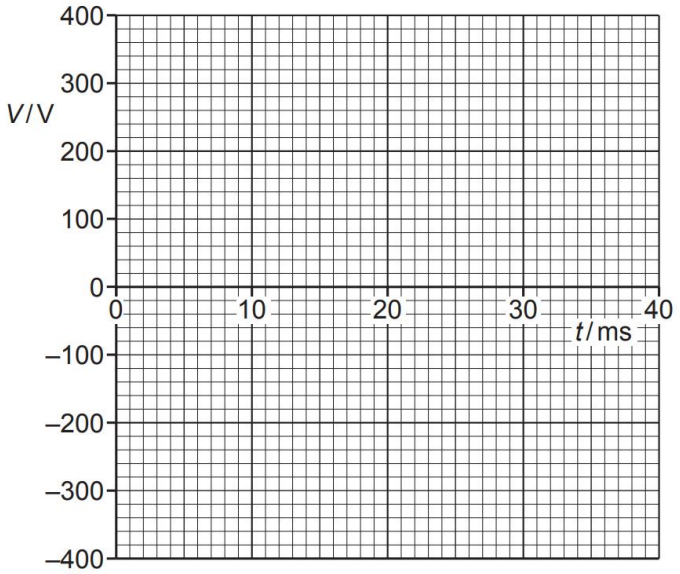
c. The power supply is connected to resistor R and a diode in the circuit shown in Fig.



i. State the name of the type of rectification produced by the diode in Fig.

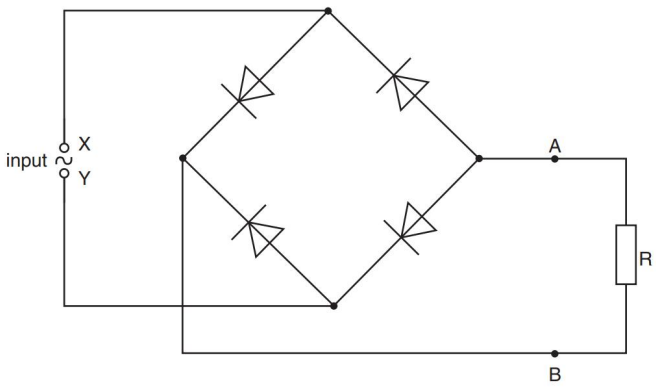
.....[1]

ii. On Fig. sketch the variation with time t of the p.d. V_R across R from time t = 0 to time t = 40 ms. [3]



iii. On Fig.1., draw the symbol for a component that may be connected to produce smoothing of VR. [1]

3. The circuit for a full-wave rectifier using four ideal diodes is shown in Fig.



A resistor R is connected across the output AB of the rectifier.

a. On Fig. 11.1.

i. draw a circle around any diodes that conduct when the terminal X of the input is positive with respect to terminal Y, [1]

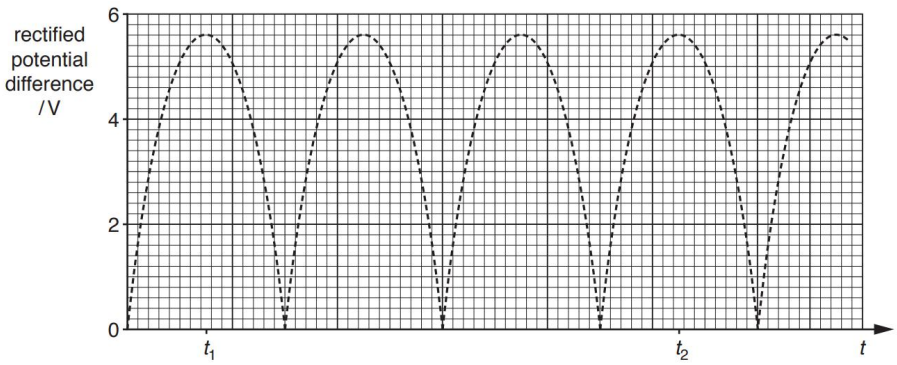
ii. label the positive (+) and the negative (–) terminals of the output AB. [1]

b. The variation with time t of the potential difference V across the input XY is given by the expression

$$V = 5.6sin380t$$

where V is measured in volts and t is measured in seconds.

The variation with time t of the rectified potential difference across the resistor R is shown in Fig.



Use the expression for the input potential difference V, or otherwise, to determine

i. the root-mean-square (r.m.s.) potential difference Vr.m.s. of the input,

$V_{r.m.s.}$ = ... V [1]

ii. the number of times per second that the rectified potential difference at the output reaches a peak value.

number = ... [2]

c. A capacitor is now connected between the terminals AB of the output.

The capacitor reduces the variation (the ripple) in the output to 1.6V.

i. On Fig., sketch the variation with time t of the smoothed output voltage for time t = t_1 to time t = t_2. [4]

ii. Suggest and explain the effect, if any, on the mean power dissipation in resistor R when the capacitor is connected between terminals AB.

.....

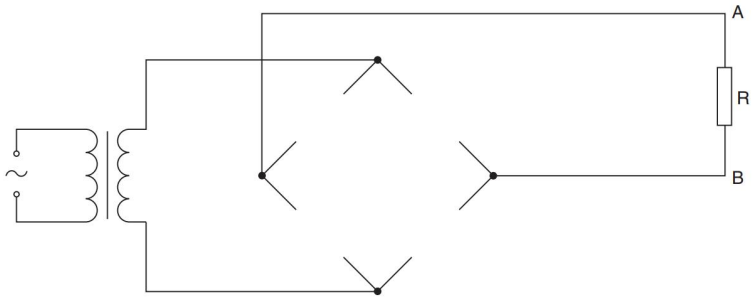
.....[2]

4. An ideal transformer has 5000 turns on its primary coil. It is to be used to convert a mains supply of 230 V r.m.s. to an alternating voltage having a peak value of 9.0 V.

a. Calculate the number of turns on the secondary coil.

number = ... [3]

b. The output from the transformer is to be full-wave rectified. Fig.4.1 shows part of the rectifier circuit

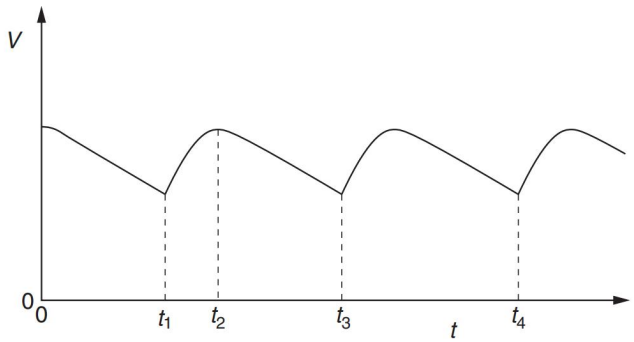


On Fig., draw

i. diode symbols to complete the diagram of the rectifier such that terminal A of the resistor R is positive with respect to terminal B, [2]

ii. the symbol for a capacitor connected to provide smoothing of the potential difference across the resistor R. [1]

c. the symbol for a capacitor connected to provide smoothing of the potential difference across the resistor R. [1]



i. State the interval of time during which the capacitor is being charged from the transformer.

from time …… to time …… [1]

ii. The resistance of the resistor R is doubled. On Fig., sketch the variation with time t of the potential difference V across the resistor. [2]